BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: MRD XI (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** Physics Paper- XI, (MRD-1) (21)

Q1. SECTION – A

Q1. Dimension of torque is.....

(A) MLT
(B) ML2T−1
(C) MLT−1
(D) ML2T−2

Q2. 1 radian is given as.....

(A) 1°
(B) 360°
(C) 57.3°
(D) 3.14°

Q3. The magnitude of vector $\vec{A} = 2\hat{i} + \hat{j} + 2\hat{k}$ is:

(A) 9
(B) 5
(C) 3
(D) 1

Q4. Torque of the body is maximum at.....

(A) 0°
(B) 30°
(C) 60°
(D) 90°

Q5. Formula for the Height of the Projectile (H) is.....

(A) $V_o^2 \sin^2\theta/2g$
(B) $V_o \sin\theta/2g$
(C) $V_o^2 \sin\theta/2g$
(D) $V_o \sin^2\theta/2g$

Q6. Unit of linear momentum is.....

(A) NS^{-1}
(B) N.m
(C) N.S
(D) $N.m^{-2}$

Q7. The expression for escape velocity is given by:

(A) $2gR_o^2$ (B) $\sqrt{2gR_o}$

(C) $gR_o^2/2$ (D) $\sqrt{gR_o}$

Q8. The example of non conservative force is.....

(A) Electric force (B) Gravitational force

(C) Frictional force (D) Magnetic force

Q9. The angular speed in radians/hours for daily rotation of our earth is.....

(A) 2π (B) 4π

(C) $\pi/6$ (D) $\pi/12$

Q10. Moment of inertia of a sphere is.....

(A) $2/5\ MR^2$ (B) MR^2

(C) $1/2\ MR^2$ (D) $5/2\ M^2R^2$

Q11. Formula for the terminal velocity is v_t =

(A) $2\rho g/9\eta$ (B) $2\rho r^2/9\eta$

(C) $2\rho gr^2/9\eta$ (D) $2\rho gr/9\eta$

Q12. According to the equation of continuity, speed of fluid through any pipe of cross sectional area is given as.....

(A) $v \propto 1/\sqrt{A}$ (B) $v \propto 1/A$

(C) $v \propto A$ (D) $v \propto \sqrt{A}$

Q13. Tuning of a radio set is an example of....

(A) Mechanical resonance (B) Musical resonance

(C) Electrical resonance (D) Free vibrations

Q14. The time period of simple pendulum is 2 seconds if its length is increased by 4 times then its period becomes.....

(A) 16 sec (B) 12 sec

(C) 8 sec (D) 4 sec

Q15. Speed of sound is 332 m/s, Newton calculated the value 281 m/s. What is the difference between the practical and theoretical value.

(A) 16% (B) 26%

(C) 06% (D) 36%

Q16. Which of the given factor has no effect on the speed of sound in a gas?

(A) Humidity (B) Pressure

(C) Temperature (D) Density

Q17. S.I unit of molar specific heat is.....

(A) J.mole.k
(B) J.mole^{-1}k
(C) Jmole^{-1}k^{-1}
(D) J.mole k

Q18. During a sunny day we see the objects in a class room even when all the electric lights are off, due to.....

(A) Reflection of light
(B) Refraction of light
(C) Diffraction of light
(D) Interference of light

Section – B

Maximum Marks: 40

Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

QII. Attempt any TEN parts. Each part carries FOUR marks.

1. How many radians account for circumference of a circle? How many steradians account for circumference of a sphere?

2. Is it possible to add three vectors of equal magnitude but different directions to get a null vector? Illustrate with diagram.

3. Explain the warning "Never use a large wrench to tighten a small bolt".

4. An aero plane while travelling horizontally, dropped a bomb when it was exactly above the target, the bomb missed the target. Explain why it missed the target?

5. What is the angle for which the maximum height reached and corresponding range are equal?

6. What is escape velocity? Briefly explain and deduce the formula.

7. Why is the acceleration of a body moving uniformly in a circle, directed towards the centre?

8. When a tractor moves with uniform velocity its heavier wheel rotates slowly than its lighter wheel, why? Explain

9. What is viscous drag? State and explain stokes law.

10. What happens to the time period of a simple pendulum if its length is doubled?

11. Briefly differentiate between transverse and longitudinal waves, write down the characteristics of the waves.

12. What is polarization of light? Discuss polarization by reflection.

13. Define first law of thermodynamics, write down the applications of first law of thermodynamics.

Section – C

Maximum Marks: 27

> Note: Attempt any THREE questions. All questions carry equal marks.

QIII. (Marks 9)

(a) [5] Explain addition of vectors by rectangular components.

(b) [4] In wedding a bullet is fired in air at speed of 500 ms^{-1} making an angle of 60° with horizontal from an AK47 rifle. (i) How high will the bullet rise? (ii) What time would it take to reach ground? (iii) How far would it go? (Ignore air resistance)

QIV. (Marks 9)

(a) [5] What is absolute potential energy? Prove that Absolute P.E = GmMe/Re

(b) [4] A child of mass 25 kg stands at the edge of rotating platform of mass 150 kg and radius 4m. The platform with the child on it rotates an angular speed of 6.2 rad/s. The child jumps off in a radial direction, what happens to the angular speed of the platform? Treat platform as a uniform disk.

QV. (Marks 9)

(a) [5] Derive mathematically the equation of continuity and relate it to the time rate of volume flow. How equation of continuity is based on conservation of mass?

(b) [4] Find the speed of sound in a steel railway track, if the density of steel is 7800 kg/m^3 and elastic modulus is 2.0×10^{11} Nm^{-2}.

QVI. Write short note on any two (2) of the following:

(a) [4.5] Damped oscillation

(b) [4.5] Young's double slit experiment.

(c) [4.5] Entropy